

Infosys Springboard Virtual Internship 7.0 Completion Report

Team Details

Batch Number : Team B

Start date : 21 July 2026

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Internship Duration: 8 Weeks

1. Project Title

AI-Based Industrial Maintenance and Process Optimization — EV Health Monitoring

2. Project Objective

The main objective of the project is to develop an AI-based industrial maintenance and process optimization system for EV charging stations. The system aims to monitor charging station parameters, analyze operational data, identify abnormal conditions, predict potential maintenance requirements, and support efficient maintenance planning. By using machine learning, data analysis, and process monitoring, the system helps improve equipment reliability, reduce unexpected downtime, and optimize the overall operation of EV charging infrastructure. .

3. Project description in detail

The project focuses on developing an AI-based industrial maintenance and process optimization solution with EV charging stations as the application domain. The system combines machine learning, database management, backend services, and a web-based monitoring dashboard to analyze the operational condition of charging stations.

The system collects and manages parameters such as temperature, voltage, current, power consumption, and other telemetry information. This data is analyzed to assess equipment health, identify abnormal operating conditions, and support predictive maintenance.

The system provides modules for charging station monitoring, telemetry, battery health, predictions, maintenance, alerts, failure history, charging sessions, operators, feedback, and analytics. The dashboard provides health indicators, visualizations, search and filtering features to help monitor the system efficiently.

The overall goal is to apply AI-based monitoring and predictive maintenance techniques to optimize industrial processes, improve equipment reliability, reduce unexpected failures, and support timely maintenance decisions

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understand EV health monitoring and project requirements	Studied the problem domain, requirements and project objectives
Week 2	Design system architecture and database	Designed system architecture and created SQLite database structure
Week 3	Develop backend and data handling	Developed backend APIs and database operations
Week 4	Develop and integrate ML component	Developed ML workflow and integrated prediction functionality
Week 5	Develop web dashboard	Developed frontend and dashboard modules
Week 6	Add advanced monitoring features	Added alerts, battery health, failure history, charging sessions and analytics
Week 7	Testing and debugging	Tested application, fixed database, API, navigation and UI issues
Week 8	Finalize project and documentation	Finalized application, documentation, GitHub repository and presentation

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Problem identification, requirements and project objectives	30 July 2026
Prototype/First Draft	Initial database, backend and monitoring interface	10 august 2026
Mid-Term Review	Core backend, ML functionality and dashboard reviewed	21 august 2026

Final Submission	Complete application and documentation finalized	4 september 2026
Presentation	Project implementation and results presented	16 september 2026

5b. Project execution details

The project was executed in multiple stages, starting with understanding the requirements and EV charging station health monitoring problem. The overall system was divided into database, backend, machine learning and frontend components.

An SQLite database was developed to store charging station, telemetry, maintenance, alerts, predictions, battery health, failure history, charging session, operator and feedback information. The backend was developed to handle application logic, database operations and API functionality.

Machine learning functionality was integrated to analyze charging station operational data and support health assessment and predictive maintenance. A web-based dashboard was then developed to display station health, telemetry, battery health, predictions, alerts, maintenance information and analytics.

The system was progressively tested and improved by fixing database, API, navigation and UI issues. Search and health-status filtering were also added to make station monitoring easier. Finally, the complete application, documentation and presentation were prepared.

6. Snapshots / Screenshots

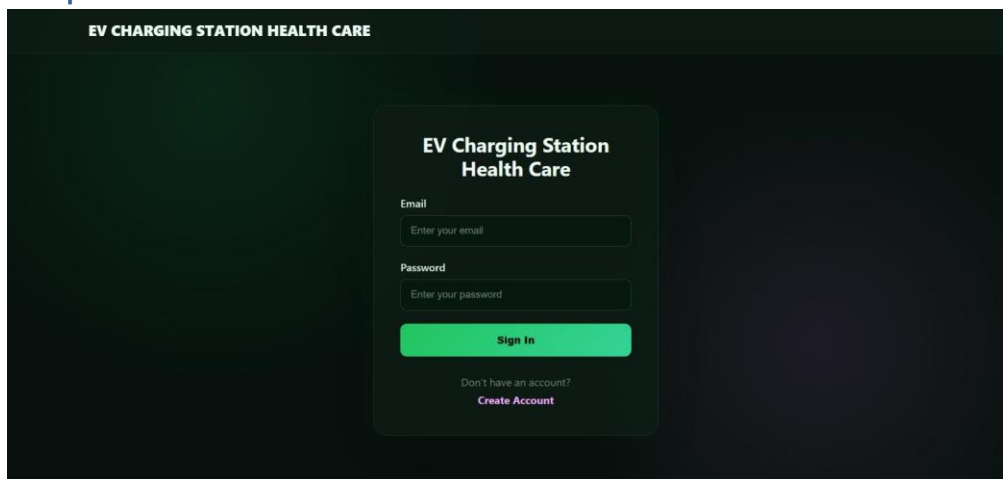


fig 1: User login Interface

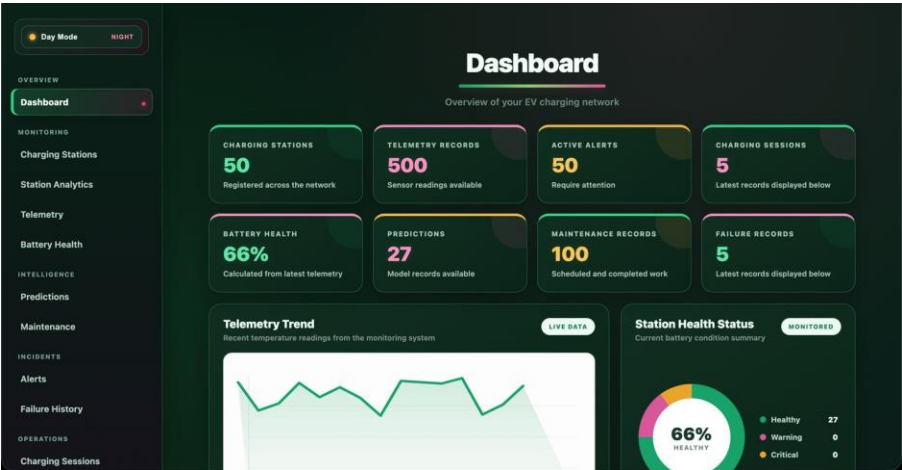


fig-2: Dashboard overview

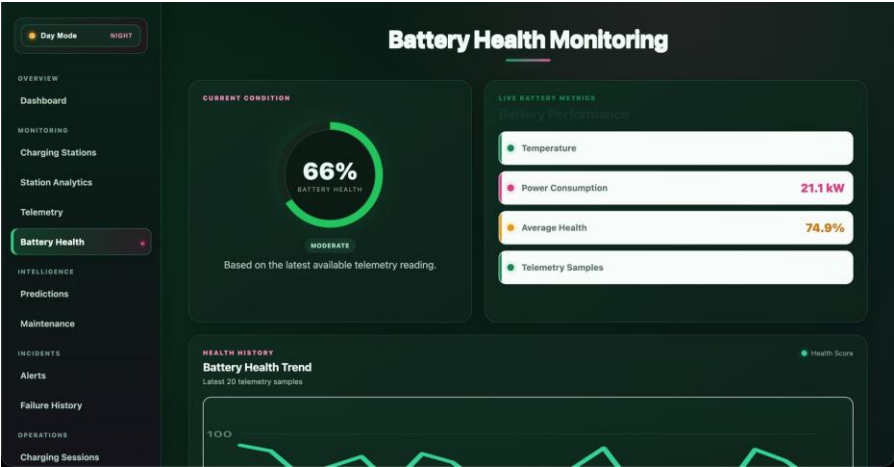


Fig 3 : Battery Health Monitoring

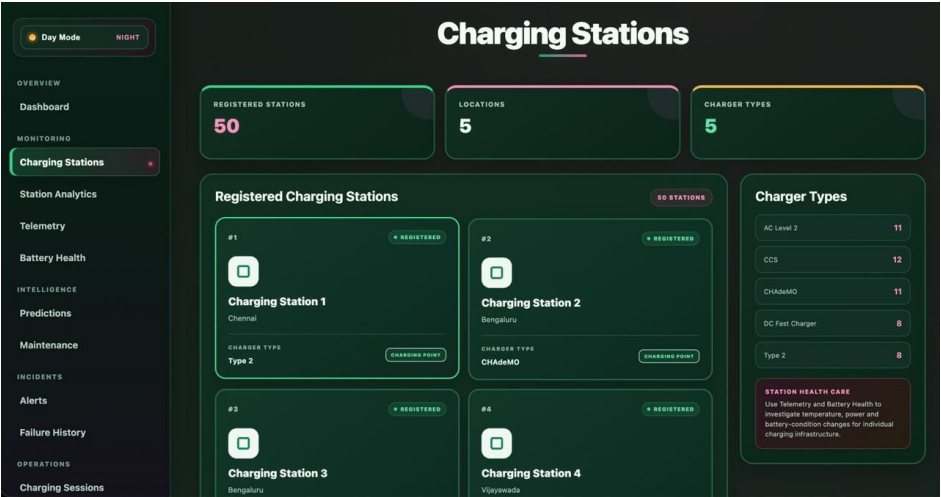


Fig 4 : Charging stations

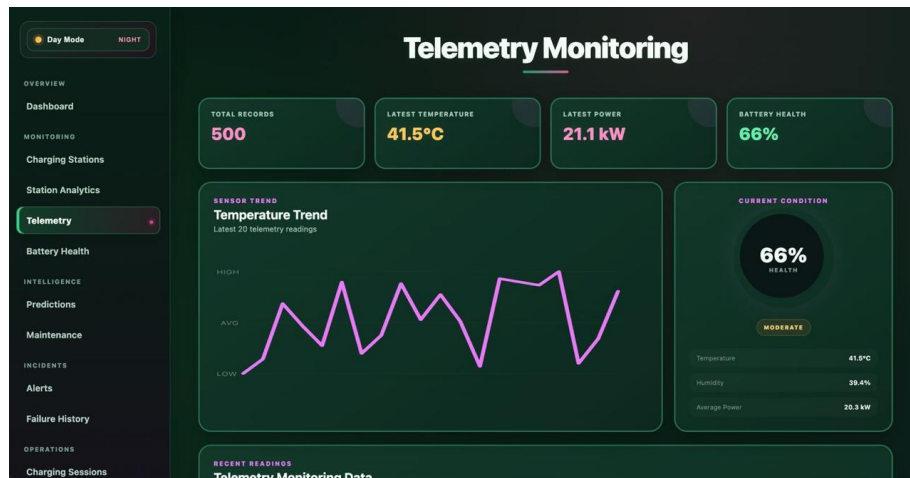


fig-5:Telemetry Monitoring

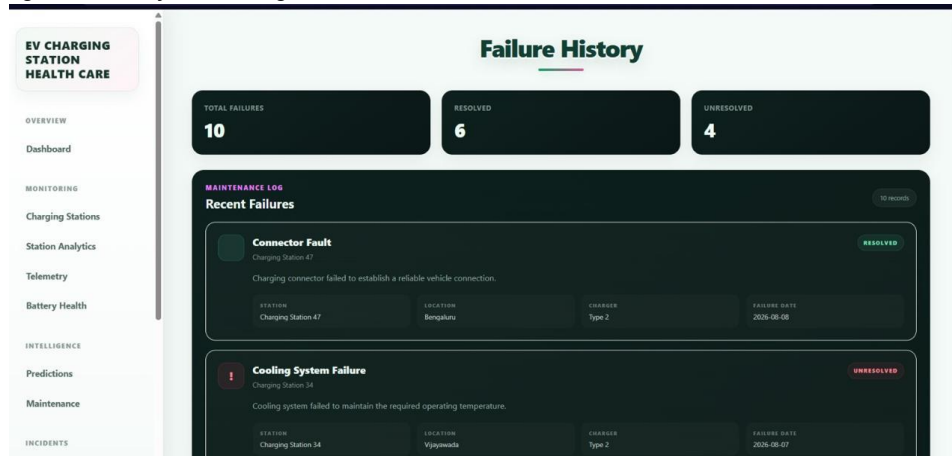


Fig-6: Failure history

7. Challenges Faced

During the development of the project, we faced challenges in integrating the different components such as the database, backend, machine learning model and frontend dashboard. Maintaining consistency between telemetry, maintenance, prediction and other modules also required careful testing.

Integrating the machine learning prediction functionality with the application and displaying the results properly on the dashboard was another challenge. We also faced issues related to database connections, API communication, navigation and frontend design.

These challenges were resolved through continuous testing, debugging, team discussions and making improvements to the application step by step. Git and GitHub were also used to manage and organize the project code.

8. Learnings & Skills Acquired

Through this internship project, I gained practical knowledge of AI-based EV health monitoring and predictive maintenance. I learned how machine learning can be applied to operational data to monitor equipment health and support maintenance decisions.

I gained experience with Python, machine learning, SQLite database management, backend/API development, web-based frontend development and data visualization. I also learned how to integrate different components into a single application and develop monitoring dashboards.

Apart from technical skills, I improved my problem-solving, debugging, teamwork, communication, documentation and presentation skills. The project helped me understand how theoretical concepts can be applied to a real-world EV infrastructure problem.

9. Testimonials from team

The project provided us with a valuable practical experience in developing an AI-based EV health monitoring system. Working on the project helped us understand the complete development process, from requirement analysis and database design to machine learning integration, backend development and frontend dashboard implementation.

The experience improved our technical knowledge, problem-solving abilities and teamwork. It also helped us understand the challenges involved in developing a real-world monitoring and predictive maintenance application.

10. Conclusion

The EV Health Monitoring and Predictive Maintenance project provided valuable practical experience in applying AI and software technologies to EV charging infrastructure. The system integrates machine learning, database management, backend services and a web-based dashboard to monitor charging station health and operational conditions.

The project provides features such as telemetry monitoring, health assessment, battery health information, predictions, alerts, maintenance records, failure history and analytics. It helped us understand how AI-based monitoring can support predictive maintenance and improve the reliability and efficiency of EV charging stations.

Overall, the internship helped bridge academic knowledge with practical implementation and improved our skills in machine learning, Python, databases, web development, debugging, teamwork and project management.

11. Acknowledgements

I would like to express my sincere gratitude to **Infosys Springboard** for providing me with the opportunity to participate in the **Virtual Internship** and work on a practical technology project.

I would also like to sincerely thank our **mentor** for the guidance, support and valuable feedback provided throughout the project. I am also grateful to my **team members** for their cooperation, teamwork and contribution during the development of the project.

This internship provided us with a valuable opportunity to apply our academic knowledge to a real-world **AI-based EV Health Monitoring and Predictive Maintenance System** and improve our technical and problem-solving skills.